



PREDICTIVE ANALYTICS PROJECT REPORT

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Abstract:

Customer attrition has always been a problem for all types of businesses. Banks and credit card companies specifically have an increased desire to keep their customers loyal. With today's data collection and modeling strategies it is possible to help any business, including banks and credit card companies, to accurately identify customers who are likely to leave. This allows the business to offer incentives to get these customers to stay loyal.

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Table of Contents

Introduction	2
Data Description.....	2
Initial EDA.....	3
Modeling	5
Feature Engineering.....	6
Feature Selection	6
Hyper-Parameter Tuning.....	6
Ensemble Model	8
Conclusion.....	8
Implications and Further Study.....	8

Introduction

Customer churn is defined as a customer who has stopped using a company's product or service. In the scope of this project a churned customer would be one that has cancelled their credit card. The goal of this project is to build a model that can correctly identify and flag customers about to churn so the company can make efforts to retain their loyalty.

Data Description

This dataset was found on Kaggle (<https://www.kaggle.com/datasets/thedevastator/predicting-credit-card-customer-attribution-with-m>) but was originally collected and uploaded to Zenodo in December 2020. The dataset contains 10,126 rows and 23 columns with no missing values. Each row represents a customer, and each column represents customer information collected within their credit card portfolio.

Column name	Type	Description
CLIENTNUM	Numerical	Unique identifier for each customer.
Attrition_Flag	Categorical	Flag indicating whether or not the customer has churned out.
Customer_Age	Numerical	Age of customer.
Gender	Categorical	Gender of customer.
Dependent_count	Numerical	Number of dependents that customer has.
Education_Level	Categorical	Education level of customer.
Marital_Status	Categorical	Marital status of customer.
Income_Category	Categorical	Income category of customer.
Card_Category	Categorical	Type of card held by customer.
Months_on_book	Numerical	How long customer has been on the books.
Total_Relationship_Count	Numerical	Total number of relationships customer has with the credit card provider.
Months_Inactive_12_mon	Numerical	Number of months customer has been inactive in the last twelve months.
Contacts_Count_12_mon	Numerical	Number of contacts customer has had in the last twelve months.
Credit_Limit	Numerical	Credit limit of customer.
Total_Revolving_Bal	Numerical	Total revolving balance of customer.
Avg_Open_To_Buy	Numerical	Average open to buy ratio of customer.
Total_Amt_Chng_Q4_Q1	Numerical	Total amount changed from quarter 4 to quarter 1.
Total_Trans_Amt	Numerical	Total transaction amount.
Total_Trans_Ct	Numerical	Total transaction count.
Total_Ct_Chng_Q4_Q1	Numerical	Total count changed from quarter 4 to quarter 1.
Avg_Utilization_Ratio	Numerical	Average utilization ratio of customer.

Initial EDA

To get a better understanding of our dataset we started by making a few count plots, histograms, and a heat map so see if we could find any noteworthy correlations or trends in the data that could benefit us in the model building process.

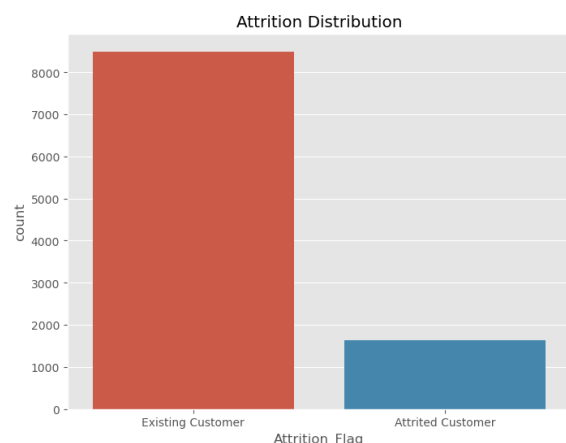


Figure 1

This data set observes about a 19% attrition rate. That means 19% of customers represented in the data set left the credit card company. This could be considered an imbalanced data set. We will need to keep in mind the recall score of our models.

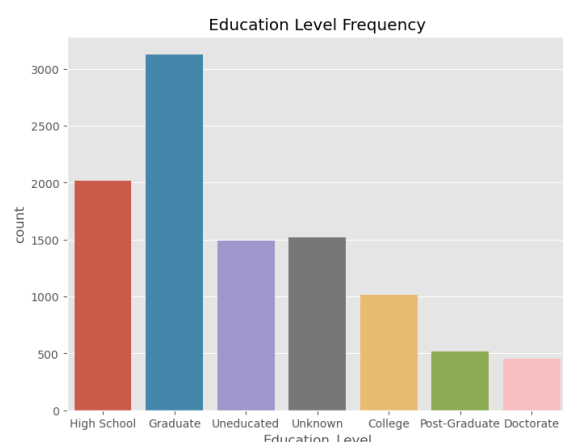


Figure 2

From the count plot we can see a majority of the customers' highest level of education is graduate followed by High School.

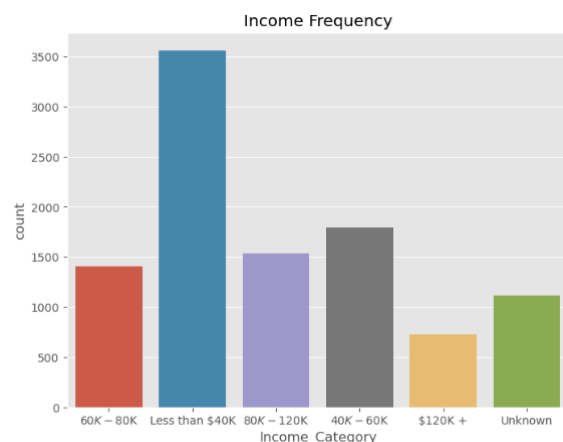


Figure 3

The count plot to the left shows that a large majority of customers earn less than \$40k per year. We can also see a decent chunk of the customers aren't reporting an income.

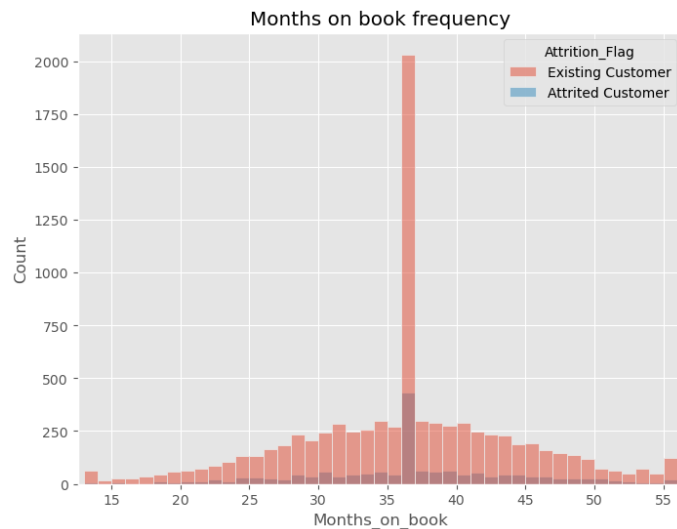


Figure 4

From the above plot we can see that most of the customers have been on books for at least 37 months. The 37 months on book is an extreme outlier being it almost half the customers. Aside from this outlier the distribution is normal.

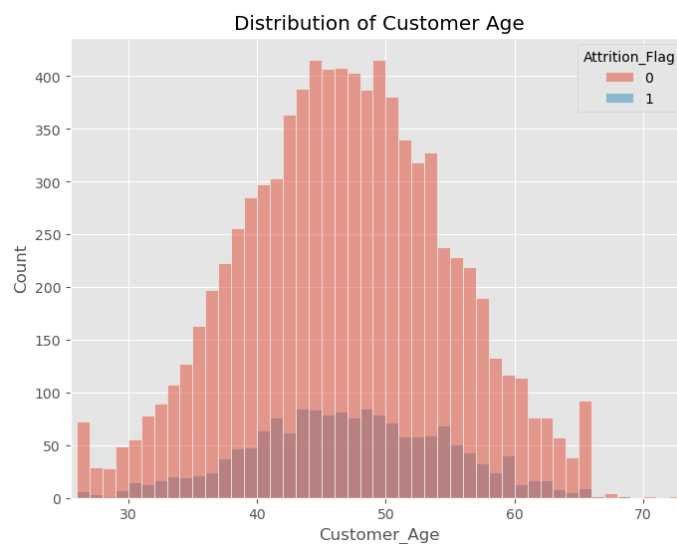


Figure 5

In the histogram to the left we can see the distribution of customer ages. This chart shows that the company's customer base is normally distributed.

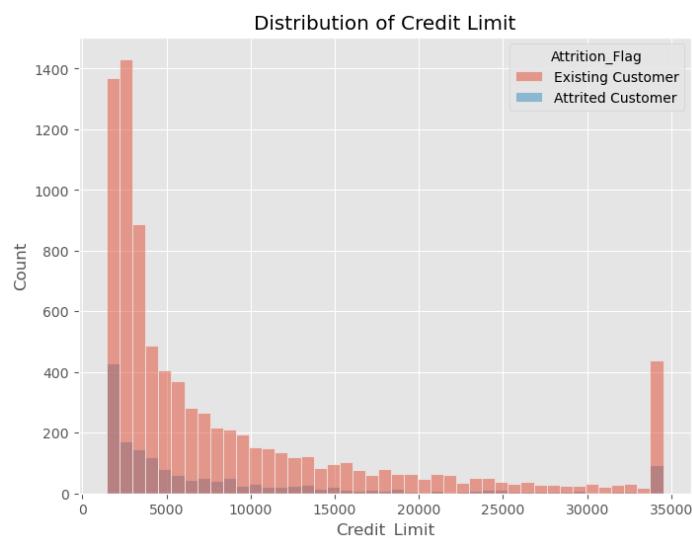


Figure 5

Here we see the distribution of the customer bases credit limit. The distribution is right skewed with a majority of customers having limits below 5,000. There is an interesting group sticking out from the trend with a limit around 34,000

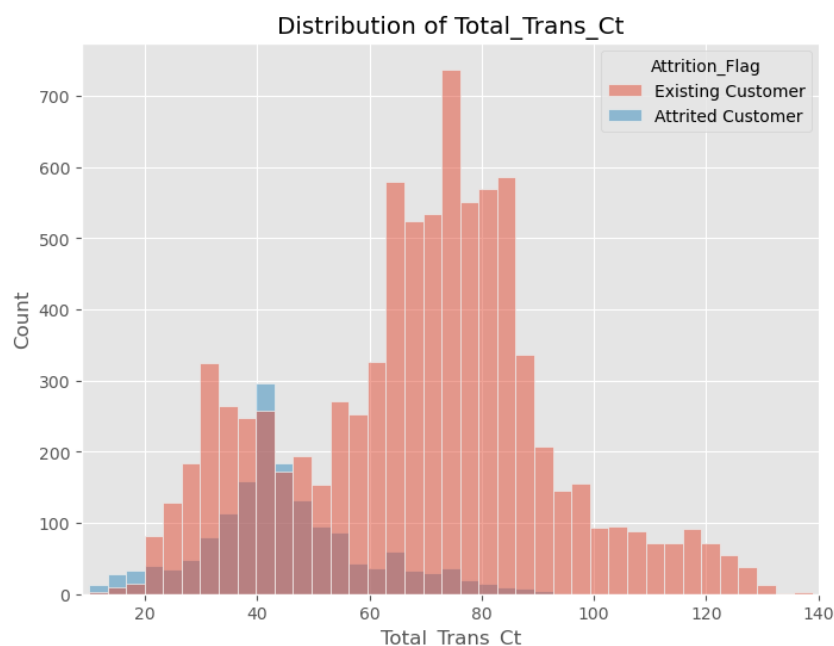


Figure 6

In the count plot to the left we can see an interesting difference in distributions between existing and attrited customers. Existing customers follow a more normal distribution while attrited customers are distributed around 40.

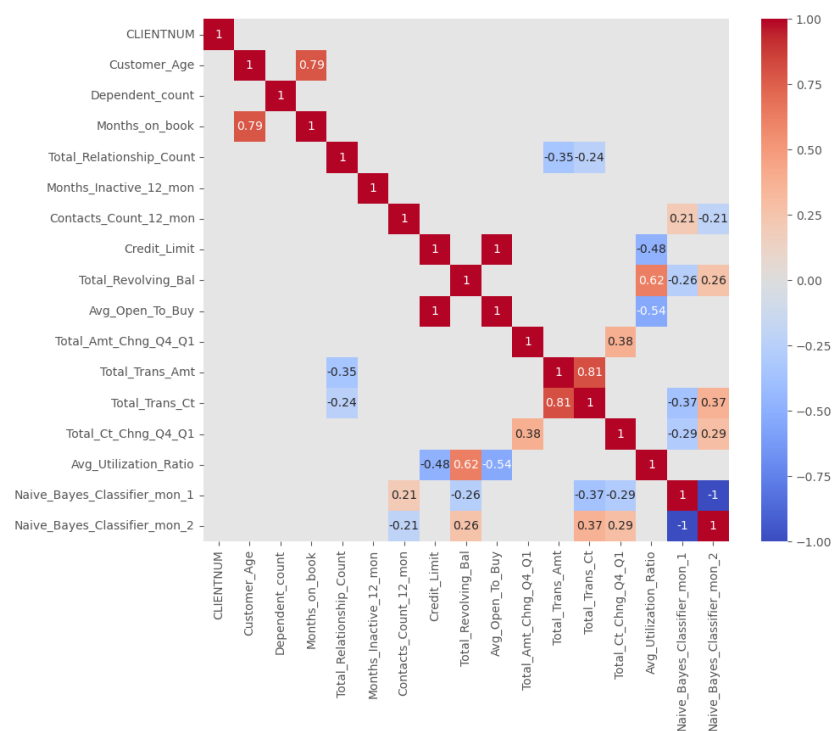


Figure 7

In this heatmap, we highlight the variables that have a negative/positive correlation. We see that there are a number of correlated variables. We could engineer a few features based on these interactions.

Modeling

The goal of this project is to build a model that can predict customer churn as accurately as possible while minimizing the false negative case. Using customer credit card portfolio information, we built a variety of models to ensure we have the optimal learners to compile into an ensemble. This section will outline our approach to reaching our final model with the best performance.

Feature Engineering

After our initial EDA we brainstormed some possible features to help boost the performance of our models. We made interaction terms based on variables which had high correlations along with flags based off our visualizations. Those features are described in the table below.

Column name	Type	Description
Interaction_1	Numerical	Customer_Age * Months_on_book
Interaction_2	Numerical	Total_Revolving_Bal * Avg_Utilization_Ratio
Interaction_3	Numerical	Total_Trans_Amt * Total_Trans_Ct
Interaction_4	Numerical	Avg_Utilization_Ratio * Avg_Open_To_Buy
Months_on_book_36_flag	Categorical	Flag for Months_on_book = 36
Credit_Limit_max_flag	Categorical	Flag for Credit_Limit = 34516
Total_Revolving_Bal_0_flag	Categorical	Flag for Total_Revolving_Bal = 0
Total_Revolving_Bal_max_flag	Categorical	Flag for Total_Revolving_Bal = 2517
Total_Trans_Ct_under_60	Categorical	Flag for Total_Trans_Ct >= 60

Feature Selection

In order to decide which features were most important in detecting churn we used RFECV with 5 folds and f1 score as the evaluation metric. We ran through this sequence 5 times and kept track of the percentage of time each feature was selected in each model and selected those used 90% of the time or higher. The following table outlines the number of optimal features for each model and their accuracy score.

Model	Number of features	Accuracy score
Random Forest	12	.9628
Ada Boost	20	.9701
Gradient Boosting	15	.9707
Hist Gradient Boost	14	.9721
XG Boost	13	.9718
Light GBM	22	.9706
Cat Boost	25	.9713

Hyper-Parameter Tuning

To find the optimal hyper-parameters for our data we used the Optuna framework to maximize the f1 score over 50 trials for each of the following models. The optimal hyper-parameter combinations for each model are listed below.

Model	Hyper-Parameters
Random Forest	n_estimators = 974 max_depth = 10 min_samples_split = 5 min_samples_leaf = 2

Ada Boost	learning_rate = 0.08 n_estimators = 438 max_depth = 3
Gradient Boosting	n_estimators = 958 max_depth = 3 min_samples_split = 8 min_samples_leaf = 6 learning_rate = 0.0547
Hist Gradient Boost	max_iter = 520 learning_rate = 0.1 l2_regularization = 0.1 min_samples_leaf = 7 max_leaf_nodes = 103 max_depth = 3 max_bins = 205
XG Boost	learning_rate = 0.02 n_estimators = 622 max_depth = 6 min_child_weight = 1 gamma = 0.003011 alpha = 3.305498 colsample_bytree = 0.51111 subsample = 0.433741
Light GBM	n_estimators = 966 learning_rate = 0.02 num_leaves = 313 min_data_in_leaf = 37 min_child_weight = 0.07766 max_depth = 9 bagging_fraction = 0.65689 feature_fraction = 0.52445 lambda_l1 = 0.9316 lambda_l2 = 1.0916
Cat Boost	iterations = 699 learning_rate = 0.1 min_data_in_leaf = 154 depth = 7 random_strength = 0.76723 bagging_temperature = 0.34078 border_count = 79 l2_leaf_reg = 64

Ensemble Model

After identifying the optimal features and hyper-parameters for our dataset we picked out our top performing models by accuracy score to ensemble, making sure there was an odd number of models so there are no ties in predictions. The following table describes the top five models and their associated accuracy scores.

Model	Accuracy score
Ada Boost	.9744
Gradient Boosting	.9736
Hist Gradient Boosting	.9734
Light GBM	.9749
Cat Boost	.9763
Ensemble model	.9827

Conclusion

In this project we were able to identify churned credit card customers. Through the use of exploratory data analysis, we familiarized ourselves with the data and were able to engineer new features that helped the performance of our models. We also used many different model optimization strategies such as hyper-parameter tuning, feature selection, and ensemble learning to reach achieve our final model with an accuracy of .98 and a recall score of .95 showing high level ability to predict the positive case with very few false negatives.

Implications and Further Study

With the high level of accuracy our model was able to produce, we believe this model, or ones very similar to it, would be very beneficial to credit card companies or companies where customer attrition is a main priority. Using the results from our model, companies would be able to flag customers that may not be fully satisfied with their services. Companies could then use that feedback to improve the customer's experience and result in a better product or service for their customers as a whole.

An area of further study we would be interested in looking at is what causes customers to churn from the company. Our dataset focuses on customers' personal data but does not have any information on credit card features customers could be dissatisfied with. If we had this type of data, we could possibly provide more information to the company on what features and services lead to higher customer satisfaction.